

SEMANTICAL ARCHAEOLOGY: A PARABLE

ABSTRACT. A somewhat fictionalized account of several interpretations of implication is presented together with comparisons between classical modal, tense, and intuitionistic logics.

Toward the middle of the last century on an obscure, uninhabited island in the eastern Mediterranean, a considerable number of clay fragments were unearthed covered with mysterious symbols. After years of careful cleaning and sorting they were finally rather surprisingly arranged to form several long panels. Even the missing parts were reconstructed by intricate measurements and clever comparisons. Nevertheless, no amount of study yielded their proper decipherment.

Late one evening a young scholar sat alone in his room poring over the photographs and drawings of the panels, pondering the question of their meaning. He had fixed on his wall several examples of the more outstanding recurrent sequences to be found in the inscriptions; in particular these:

- (1) $\alpha\alpha\beta\alpha CCC,$
- (2) $\alpha\gamma C\alpha\gamma C\beta\alpha\gamma CCCC,$
- (3) $\gamma\alpha C\gamma\beta CCC\beta\alpha CC.$

We need not retell here the exciting story, which figured so heavily in all the news reports, of how he came *first* to reverse the order of the symbols, *then* to make the very ingenious substitution of $\supset$ for C and p, q, r for α, β, γ , and *finally* to discover the necessary, though far from obvious, change of word order which produced:

- (1') $[[[p \supset q] \supset p] \supset p],$
- (2') $[[[[[r \supset p] \supset q] \supset [r \supset p]] \supset [r \supset p]],$
- (3') $[[p \supset q] \supset [[q \supset r] \supset [p \supset r]]].$

Rather, we shall be more interested in this paper in the less dramatic problem of *interpretation*.

It goes without saying that the first idea occurring to everyone (after

the most difficult part of the task had already been done) was that these curious sequences of symbols represented *logical truths*. This was an especially appealing hypothesis in view of the well-known traits of a related and still existing people on a nearby island. They had preserved over the centuries a rudimentary logic and had indeed asserted the general correctness of principles similar to (1)–(3). This fact was noted for the first time by an anthropological field worker, who lived for some years among the tribe, a warlike and unpleasant group. One day he had occasion to enquire of his informant just what he meant by *implication*. The man at once became astonishingly angry and started to jump around excitedly, shouting: “What do you mean, what do I mean? Implication *is* implication! Have you no concept of the *true* and the *false*?” Somehow it was clear that the anthropologist had stumbled upon a highly touchy point of tribal religion, but the heated retort did serve to communicate some clue to the unhappy investigator. As he was later able to show in his thesis, this hard-headed philosophy of the tribe which demanded that every statement must either be true or false leads directly to a logic that can be interpreted by *truth tables* – or at least he showed that the interpretation agreed most favorably with the available data. Furthermore, he thought he was even able to identify occurrences of truth tables in some of the more elaborate punched-card work of the tribe, but unfortunately the manuals describing this handicraft were never shown to outsiders.

It is such a pity that the young man who originally deciphered the panels died so early (from lack of central heating some say) before he had a chance to complete his labors. We are, however, able to follow his reasoning about the proper interpretation of the inscriptions quite well from the notebooks he left. It seems that the island people (already extinct in classical times) were well known from legends to be sweet, reasonable, tolerant, and highly theoretical – the apparent causes of their downfall. Before the disastrous wars, they had travelled widely and collected vast quantities of information. The site where the panels were found was apparently (and this was reasonably confirmed by later digs) the *university library*. The panels themselves were decorative in nature and obviously commemorated their outstanding achievements in pure logic. What our young scholar noted at once was that examples of both *valid* and *invalid* principles were given. Thus (2') and (3') occurred on the positive side, while (1') was on the negative side. This discovery, of course, ruled

out the straightforward truth-table interpretation familiar from the neighboring warlike tribe leaving a serious puzzle.

With a brilliant stroke of insight our scholar suggested that by virtue of the islanders' interests it was most likely that they had developed a theory of *universal implication* applicable to various situations. He thus took as an hypothesis the underlying notion of *situation* and demonstrated how certain laws of material implication *fail* for the corresponding strict implication. And this was his remarkable explanation of the panels.

As you can imagine this piece of work was considered a paradigm of scholarship for a long time, but sad to relate it was eventually *disproved*. You see, the reconstructions of the missing parts of the panels were too *conjectural*. Thus, after some more fragments were unearthed, the formula (2) mentioned above was found to have been actually on the *negative* side. Further investigation also showed that the two formulae:

$$(4) \quad \alpha\alpha C\beta C$$

$$(5) \quad \alpha\beta C\alpha C$$

were respectively positive and negative. Now this last finding (which, by the way, would be written:

$$(4') \quad [q \supset [p \supset p]],$$

$$(5') \quad [p \supset [q \supset p]],$$

in modern notation) did not at all conflict with the famous interpretation, but the fact about (2) was *fatal* to the theory. The sensation caused by the announcement of this discovery at the international congress will long be remembered.

It will probably be helpful at this point to spell out some of the details of this interpretation. We recall from two-valued logic the well-known method of Venn diagrams. We specify first a certain convenient universal set V and interpret a proposition p as corresponding to a subset $P \subseteq V$, the set of all elements of V for which the proposition holds. The usual connectives, then, correspond to simple operations on sets; in particular, the material implication $[p \supset q]$ corresponds to the set-theoretical combination

$$(V \sim P) \cup Q.$$

Under this correspondence, a *valid* compound proposition is one whose corresponding set-theoretical combination works out to be V no matter

which sets $P, Q, \dots$, etc. are used for the interpretation of the atomic propositions $p, q \dots$.

So much for material implication. The universal implication we have in mind for explaining the panels is defined quite differently. Supposing the elements of V to represent situations, we say that an implication $[p \supset q]$ can be regarded as true in the *present* situation if, and only if, for *arbitrary* situations in which p is true, q is always true. In other words, we must take a more general, less provincial view of the meaning of implication. More precisely, in terms of the corresponding sets we define an operation $\Rightarrow$ as follows:

$$(P \Rightarrow Q) = \begin{cases} V & \text{if } P \subseteq Q, \\ \emptyset & \text{otherwise,} \end{cases}$$

where $\emptyset$ denotes the empty subset of V . It is the strict use of $\emptyset$ that distinguishes this notion from material implication. One may now check in detail that (2), (3), (4) are valid under this interpretation, while (1) and (5) are not.¹

We need not catalogue here the endless list of artificial attempts to save the interpretation, for, several months later, a most satisfactory solution was proposed by one of the outstanding Japanese scholars in the field. He reminded us of the danger in giving a too 'Western' interpretation to the islanders' view of logic. (Note for example that they wrote from right to left!) He demonstrated by reference to the many sundials that had been newly dug up their overriding preoccupation with *time*. Moreover, he showed beyond a doubt that the site where the panels were discovered was a *grammar school* (in the older sense of the term) and *not* a university library at all. With this refined information at his disposal he was able to put forward a new interpretation: the meaning of implication had nothing to do with varying situations but was influenced by ideas of *tense*. More specifically, as the islanders were for the most part easy going, it seemed reasonable to read the implication in the *present progressive tense*. We might say something such as: *it is being the case that p implies q*, but it is rather difficult to translate into a western tongue like English.

Inasmuch as this informal description of the interpretation may not be too clear, I should like to interpose here the mathematical formulation. The change in point of view over the previous interpretation has replaced the rather unknown set V by the set T of *times* (which we can surely iden-

tify with the set of *all* real numbers.) Now we imagine the truth value of a proposition varying as a function of time $t \in T$, and then instead of functions we represent (the truth-value meaning of) a proposition by the subset $P \subseteq T$ of times at which the proposition p is true. The interpretation of implication (which we shall again write as $(P \Rightarrow Q)$ for lack of a better symbol) can be given as follows: $t \in (P \Rightarrow Q)$ (that is, *the implication is true at time t*) if and only if there is some nonzero interval I around t , such that *for all $t' \in I$, if $t' \in P$, then $t' \in Q$* . That this implication is related to the previous universal implication is clear, but it is certainly more subtle. By an *interval around t* we can understand the set of all t' with $s \leq t' \leq u$, where s and u are real numbers such that $s < t < u$. The reason for calling this tensed implication *progressive* is that, for it to be true at a certain time, the ordinary material implication must be progressively true starting from some time in the past and going through to some time in the future. It is not so difficult to prove that formulae (3) and (4) are tautological (in the new sense, of course); while (1), (2) and (5) are not.

Despite the charming ingenuity and well-reasoned character of the Japanese hypothesis, I find it now my sober duty to report to you that it is wrong – or at least requires extensive revision. The modern techniques of radioactive dating have shown that the panels were inadvertently *mis-assembled*. After rearrangement of some unfortunately interchangeable parts, I can tell you from my own examination of them in the museum that (5) must definitely be counted on the *positive* side. So again a seemingly water-tight theory is discredited.

If you care to try to rescue the Japanese interpretation, let me call to your attention some other facts that have recently come to light. Excavation at the temple has disclosed an underground storehouse containing religious scrolls amazingly well preserved in sealed urns. It seems that at a certain period there was a crisis in religious belief causing wide-spread dissatisfaction due, no doubt, to the mounting pressures from aggressive enemies. In order to save the situation and to justify his foreign policy, the king (who was also the high priest) had a divine revelation: *there is no beginning and no end*. All the ancient sacred texts were at once sealed up in the urns and new ones were speedily supplied. Fortunately for us the scribe who was responsible for the sealed files wrote in his spare time a history of the crisis, which he also deposited in one of the urns.

Now you may have difficulty in appreciating the influence of this change

in dogma on logic, but the impact would have been very soon felt in the grammar schools. In the case of our panels – so the theory goes – some of them had to be taken down and replaced by fresh ones, because formerly valid principles could no longer be considered sound. (Note that this accounts very nicely for the difference in dates of the panels.) Now in order not to shock the people too much, the transition would have had to have been prepared rather carefully. In the case of logical reasoning the following fascinating point was made: *nothing, after all, really happens instantaneously*. Thus the new dogma is not all that surprising. To use a modern example: when you turn on the electric light, you may *think* it lights instantaneously, but it does not. It takes a moment for the filament to warm up. In turning the switch, you may *think* you did it instantaneously, but you did not. It took some turning to get it to move. And so on. Thus, since the propositions that apply to important matters are of a *positive character* (something *is* burning, something *is* turning, and the like), it follows that at any moment at which they are true there will be a whole *interval* around that moment at all times of which the same proposition is also true. This is the correct scientific explanation of the dogma about no beginning and no end. One may now check for himself that of (1)–(5), when they are applied just to the *dogmatically meaningful* propositions, only (3), (4), (5) are valid, while (1) and (2) are not. In this way the tense-logic hypothesis is preserved. (I am sorry that I do not have the space here to show how the dogma was applied to foreign policy.)

Workable as it is, I cannot express personal satisfaction with the revised Japanese interpretation. Maybe it is just too intricate to be convincing. My own hypothesis is quite different: We should not forget that according to the otherwise reliable heroic epics, the islanders originally came from the mainland. Left behind were half the tribe who after many hardships found their way to the far northwest coast. Toughened by their trials, they did not die out at all but eventually became the leading business people of the area. This is most fortunate for historical research, because their records are much better preserved than those of the extinct islanders. This makes it possible to reconstruct many of the beliefs and attitude they carried with them from their homeland.

One aspect of their ancient philosophy that has always impressed me is embodied in the saying: *if you tell me something can be done, you've*

got to show me how (as rendered in my own rough translation). It may be that the generally carefree existence of the islanders tended to deemphasize such ideals, but I have found many similar references in the urn scrolls. Now my theory has it that such attitudes have a thoroughgoing effect on one's logic, and I shall now attempt to produce thereby a reconstruction of the meaning of implication. Consider the problem of establishing $[p \supset q]$. There may be many ways of doing it. At the moment we may not know whether p is true, but we do have some idea of the many ways in which its truth might be established. If we are going to claim that $[p \supset q]$ is correct, we had best exhibit a definite *connection* from p to q – otherwise we shall never be able to convince any of these tribesmen. What to do? Well, why not produce a method that *transforms* any means of establishing p into a means of establishing q . That will certainly convince everyone.

A mathematical formulation can also be supplied for this interpretation. We call a means of establishing a proposition a *construction* for that proposition, and associate with the proposition the *set* of all constructions which establish it. Suppose P is associated with p , and Q with q . What will be associated with $[p \supset q]$? (Again let us call the set $(P \Rightarrow Q)$.) In view of the above intuitive discussion, we could try to define $(P \Rightarrow Q)$ as the set of *all* functions defined on P taking values in Q . (In mathematical notation this is often written as Q^P .)

By way of example consider the principle of the syllogism (3). Now the set

$$S = ((P \Rightarrow Q) \Rightarrow ((Q \Rightarrow R) \Rightarrow (P \Rightarrow R)))$$

seems somewhat complicated, but it can be easily analyzed. Suppose $f \in (P \Rightarrow Q)$, so we know it is a function. We wish to transform it into something we might call

$$\tau(f) \in ((Q \Rightarrow R) \Rightarrow (P \Rightarrow R)).$$

How to define this $\tau(f)$? Well, it must be a function. Suppose $g \in (Q \Rightarrow R)$. We must then have

$$\tau(f)(g) \in (P \Rightarrow R)$$

But how to define $\tau(f)(g)$? Well again, it must be a function. Suppose $x \in P$. We must finally have

$$\tau(f)(g)(x) \in R.$$

How to define it? Easy:

$$\tau(f)(g)(x) = g(f(x)),$$

as anyone can see. Since we know the sets over which f, g, x range, the above equation defines $\tau \in S$ rigorously. This example demonstrates the naturalness of the interpretation.

What is significant in the example of the syllogism is that $\tau \in S$ was defined *without* reference to the characters of the elements of P, Q, R . That is, we established (3) according to the interpretation on the basis of *form* alone. That is why it is a valid principle of logic. Similarly (and more simply) we can establish (4) and (5). But this is *not* the case with (1) and (2). To reject (1), take Q as a *two*-element set and P as a *five*-element set which results by adjoining *three* new elements to Q ; thus $Q \subseteq P$. Consider a mapping $f \in (P \Rightarrow Q)$. Now *two* of the three new elements of P must map into the *same* element of Q . This rule defines a function $F \in ((P \Rightarrow Q) \Rightarrow P)$ where $F(f)$ is the element of Q just described. Note that this function F was defined with reference only to the *division* of P into its two parts (viz. Q and $P \sim Q$.) Hence, the function F is *invariant* under permutations of the elements *within* the two parts. Now suppose we had a general method of establishing (1). Then we would have a function

$$\tau \in (((P \Rightarrow Q) \Rightarrow P) \Rightarrow P)$$

invariant under *all* permutations of P and Q (without reference to the parts). Therefore $\tau(F)$ would be invariant (at least) with respect to permutations that *do* respect the division into parts. But consider that either $\tau(F) \in Q$ or $\tau(F) \in P \sim Q$. The two-element part and the three-element part can be permuted *at will*. Thus $\tau(F)$ is *not* invariant. In this way (1) is shown to be *invalid*. You may feel that this argument is too complicated, but if I may, I wish to remind you of the fallacy of considering the pre-classical peoples incapable of reason. Nor should one be misled by the quaint forms into which they sometimes cast their proofs.

Needless to say, I feel that my own interpretation has a certain purity not found in the Japanese interpretation. But I would be dishonest if I did not tell you that it has been *proved* that the principles valid in the two interpretations are *exactly the same*.² In other words, no mere collection of data can ever distinguish between them in the same way we rejected

the older interpretations. Until we find the missing grammar books that explain exactly how the islanders read their implications, the two interpretations, based on such different approaches, will have to remain hypothetical.

In conclusion it may be of interest to note that a school of thought has recently sprung up which considers *all* of (1), (2), (4) and (5) invalid and only permits (3)! The multifarious historical and psychological considerations that have led to this remarkable opinion are too complex to recount here. It may be remarked, however, that *no* substantial *semantical* evidence has ever been produced in support of this hypothesis, a circumstance which detracts considerably from an otherwise carefully developed theory.³ Therefore it does not seem warranted at the moment to doubt the reasonableness of the conclusions reported here. Nevertheless it must always be borne in mind that future revelations may require a complete reassessment.

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REFERENCES

An earlier version of this paper was read at a symposium sponsored by the Olivetti Company in Milan, Italy, in October 1968. The proceedings of that symposium, *Linguaggi nella società e nella tecnica*, Edizione di Comunità, Milan 1970, contain a longer paper from which the present one is extracted, 'The Problem of Giving Precise Semantics for Formal Languages'.

* It was through a very curious chain of circumstances that the present manuscript came into my possession. I am afraid that at the moment I am not at liberty to reveal its true authorship. Hence, if one wishes, he may regard it purely as a piece of fiction. This explains the title: I have called it a *parable* under the somewhat old fashioned idea (which is fast losing ground these days) that even a work of fiction can convey a message. The original title of the paper was, simply: 'Semantical Archaeology'.

¹ For more information on this and other kinds of implication, see A. N. Prior, *Formal Logic*, 2nd ed., Clarendon Press, Oxford, 1962.

² For this result one must combine theorems of A. Tarski and H. Läuchli. Cf. A. Tarski, 'Der Aussagenkalkül und die Topologie', *Fundamenta Mathematicae* 31 (1938) 103–134; reprinted in translation in *Logic, Semantics, Mathematics* (ed. by J. H. Woodger), Clarendon Press, Oxford, 1955. Also, H. Läuchli, 'An Abstract Notion of Realizability for Which Intuitionistic Predicate Calculus Is Complete' to appear in the proceedings of the Buffalo Conference 1968. The idea of the interpretation through transformations is due to Läuchli though it derives from the well-known work by Kleene on intuitionism.

³ For more details and references see A. R. Anderson, 'Some Open Problems Concerning the System E of Entailment', *Acta Philosophica Fennica* 16 (1963) 7–18.